

Problem Set 2 - IO

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Question 1

Exploring the Data

```
# Load necessary libraries  
library(dplyr)
```

Attaching package: 'dplyr'

The following objects are masked from 'package:stats':

filter, lag

The following objects are masked from 'package:base':

intersect, setdiff, setequal, union

```
library(ggplot2)  
library(readr)  
library(lubridate)
```

Attaching package: 'lubridate'

The following objects are masked from 'package:base':

date, intersect, setdiff, union

```
library(here)
```

here() starts at /Users/tate/SchoolWork

```
# Load the dataset
data <- read_csv(here("Y2S1/I0/ps2/data", "prod_data_all.csv"))
```

Rows: 57748 Columns: 9

```
-- Column specification -----
Delimiter: ","
chr (2): industry, country
dbl (7): id, year, Y, Y2, L, K, M
```

i Use `spec()` to retrieve the full column specification for this data.
i Specify the column types or set `show\_col\_types = FALSE` to quiet this message.

```
# Clean dataset to harmonize year discrepancy - 1981 = 81
data <- data %>%
  mutate(year = ifelse(year < 100, year + 1900, year))

# Display the first few rows of the dataset
head(data)
```

```
# A tibble: 6 x 9
   id year    Y    Y2    L    K    M industry country
  <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <chr>    <chr>
1 10250 1981 296. 113. 2.86 15.5 183. metal    col
2 10732 1989 4733. 2566. 86.2 470. 2167. metal    col
3 10732 1990 4800. 2608. 73.6 644. 2192. metal    col
4 10962 1982 7090. 1836. 37.9 5538. 5254. metal    col
5 10962 1983 4942. 1343. 40.5 6314. 3599. metal    col
6 11192 1981 1813. 1311. 24.0 1723. 502. metal    col
```

1.1

```
# Summary Statistics of the data
data %>%
  group_by(country, industry) %>%
  summarize(
    total_obs_sect = n_distinct(id),
    .groups = 'drop'
  ) %>%
  ungroup() %>%
  arrange(country, industry)
```

```
# A tibble: 10 x 3
  country industry total_obs_sect
  <chr>    <chr>         <int>
1 chile   apparel         856
2 chile   food           2584
3 chile   metal          1038
4 chile   text            836
5 chile   wood            867
6 col     apparel         744
7 col     food            908
8 col     metal           632
9 col     text            475
10 col    wood            173
```

```
data %>%
  group_by(year, industry) %>%
  summarize(
    n_obs_t = n_distinct(id),
    .groups = 'drop'
  ) %>%
  ungroup() %>%
  arrange(year, industry) %>%
  print()
```

```
# A tibble: 90 x 3
  year industry n_obs_t
  <dbl> <chr>      <int>
1  1979 apparel     354
2  1979 food      1171
3  1979 metal     373
4  1979 text      391
```

```

5 1979 wood      340
6 1980 apparel   317
7 1980 food     1067
8 1980 metal     350
9 1980 text      356
10 1980 wood     306
# i 80 more rows

```

Chile: 1. Food - 2584, 2. Metal - 1038, 3. Wood - 867, 4. Apparel - 856, 5. Textile - 836

Colombia: 1. Food - 908, 2. Apparel - 744, 3. Metal - 632, 4. Textile - 475, 5. Wood - 173

Apparel Over Time - 1979: 354 → 1996: 292 - Peak at 1981 - 979

Food Over Time - 1979: 1171 → 1996: 1178 - Peak at 1981 - 1951

Metal Over Time - 1979: 373 → 1996: 443 - Peak at 1981 - 890

Textiles Over Time - 1979: 391 → 1996: 304 - Peak at 1981 - 752

Wood Over Time - 1979: 340 → 1996: 290 - Peak at 1981 - 465

1.2

```

# Creation of Investment Variable with 10 per depreciation rate
data <- data %>%
  arrange(id, year) %>%
  group_by(id) %>%
  mutate(
    dep      = 0.10,
    k_next   = lead(K),
    i_raw    = k_next - (1 - dep) * K,
    log_i    = ifelse(i_raw > 0, log(i_raw), NA_real_),
    log_L    = ifelse(L > 0, log(L), NA_real_),
    log_K    = ifelse(K > 0, log(K), NA_real_),
    log_Y2   = ifelse(Y2 > 0, log(Y2), NA_real_)
  ) %>%
  ungroup()

```

Warning: There were 2813 warnings in `mutate()`.

The first warning was:

i In argument: `log\_i = ifelse(i\_raw > 0, log(i\_raw), NA\_real\_)`.

i In group 3: `id = 54`.

Caused by warning in `log()`:

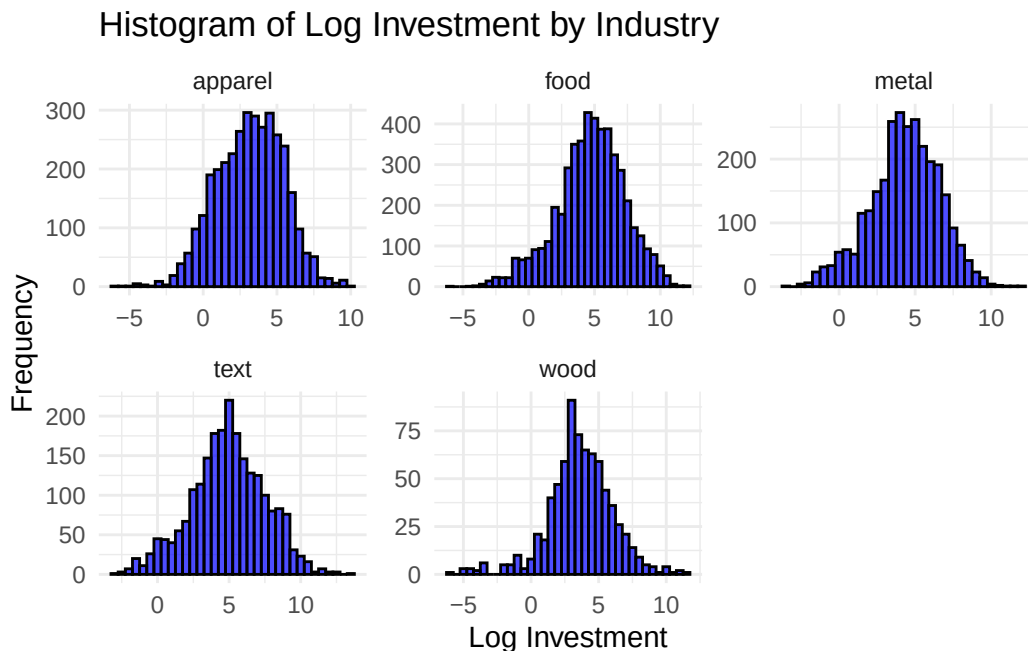
! NaNs produced

i Run `dplyr::last\_dplyr\_warnings()` to see the 2812 remaining warnings.

```
data_col <- data %>% filter(country == "col")
data_chi <- data %>% filter(country == "chile")

# Visualize with Histogram for each industry
ggplot(data_col, aes(x = log_i)) +
  geom_histogram(binwidth = 0.5, fill = "blue", color = "black", alpha = 0.7) +
  facet_wrap(~ industry, scales = "free") +
  labs(title = "Histogram of Log Investment by Industry",
       x = "Log Investment",
       y = "Frequency") +
  theme_minimal()
```

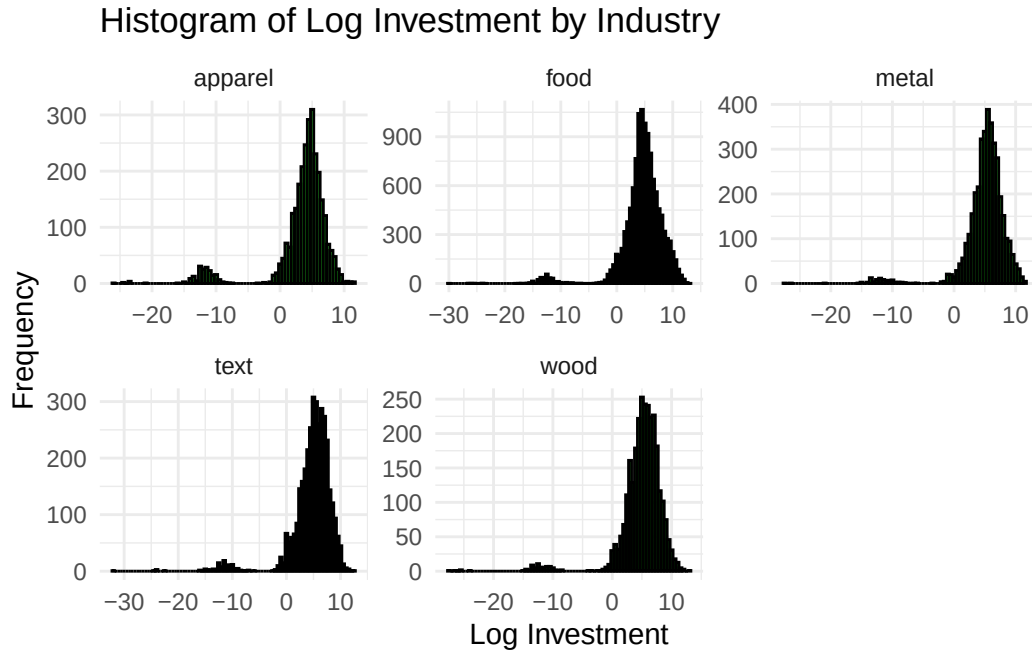
Warning: Removed 3768 rows containing non-finite outside the scale range
(`stat\_bin()`).



```
ggplot(data_chi, aes(x = log_i)) +
  geom_histogram(binwidth = 0.5, fill = "green", color = "black", alpha = 0.7) +
  facet_wrap(~ industry, scales = "free") +
```

```
labs(title = "Histogram of Log Investment by Industry",
     x = "Log Investment",
     y = "Frequency") +
theme_minimal()
```

Warning: Removed 14540 rows containing non-finite outside the scale range (`stat\_bin()`).



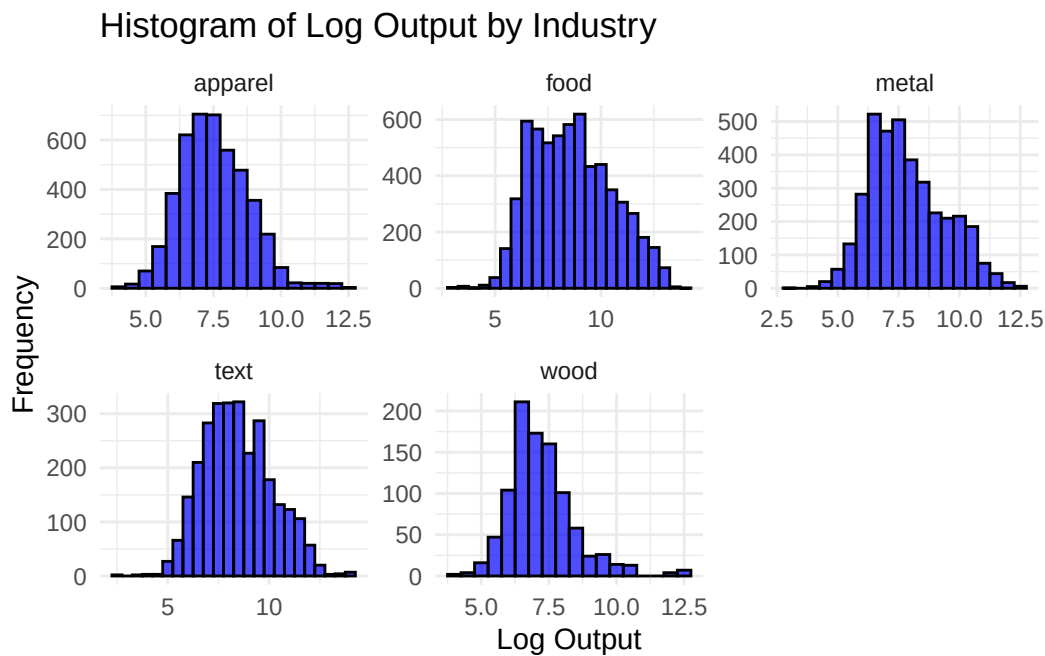
From these graphs, it seems that most industries have similar levels of investment, though textiles, wood, and metal appear to have slightly higher investment levels on average, while apparel is lower.

1.3

```
data_col <- data_col %>%
  group_by(industry) %>%
  mutate(
    log_Y = log(Y),
  ) %>%
  ungroup()
```

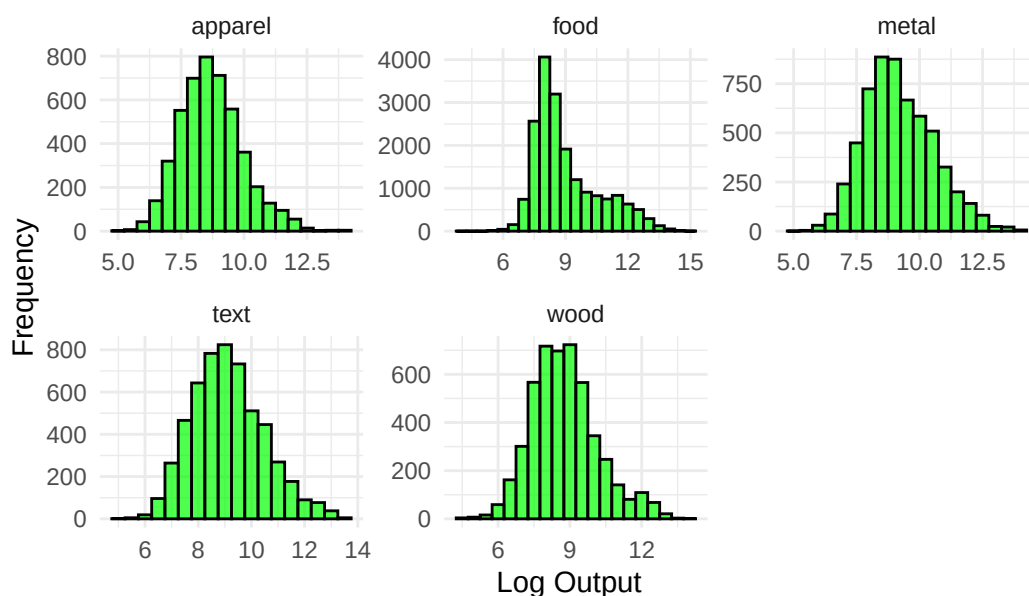
```
data_chi <- data_chi %>%
  group_by(industry) %>%
  mutate(
    log_Y = log(Y),
  ) %>%
  ungroup()

# Visualize with Histogram for each industry
ggplot(data_chi, aes(x = log_Y)) +
  geom_histogram(binwidth = 0.5, fill = "blue", color = "black", alpha = 0.7) +
  facet_wrap(~ industry, scales = "free") +
  labs(title = "Histogram of Log Output by Industry",
       x = "Log Output",
       y = "Frequency") +
  theme_minimal()
```



```
ggplot(data_chi, aes(x = log_Y)) +
  geom_histogram(binwidth = 0.5, fill = "green", color = "black", alpha = 0.7) +
  facet_wrap(~ industry, scales = "free") +
  labs(title = "Histogram of Log Output by Industry",
       x = "Log Output",
       y = "Frequency") +
  theme_minimal()
```

Histogram of Log Output by Industry



1.4

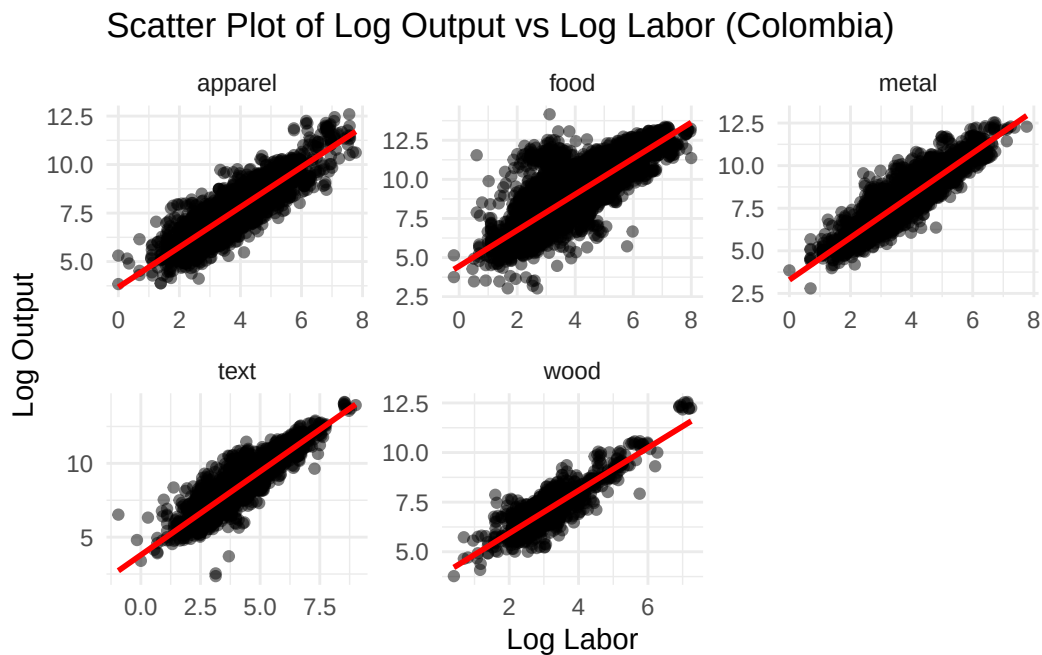
```
# Scatter plot of log_Y and log_L for each sector in each country
data_col <- data_col %>%
  group_by(industry) %>%
  mutate(
    log_L = log(L),
    log_K_ind_col = log(K),
    log_L_ind_col = log(L)
  ) %>%
  ungroup()
data_chi <- data_chi %>%
  group_by(industry) %>%
  mutate(
    log_L = log(L),
    log_K_ind_chi = log(K),
    log_L_ind_chi = log(L)
  ) %>%
  ungroup()

ggplot(data_col, aes(x = log_L, y = log_Y)) +
  geom_point(alpha = 0.5) +
```



```
geom_smooth(method = "lm", color = "red") +
facet_wrap(~ industry, scales = "free") +
labs(title = "Scatter Plot of Log Output vs Log Labor (Colombia)",
      x = "Log Labor",
      y = "Log Output") +
theme_minimal()
```

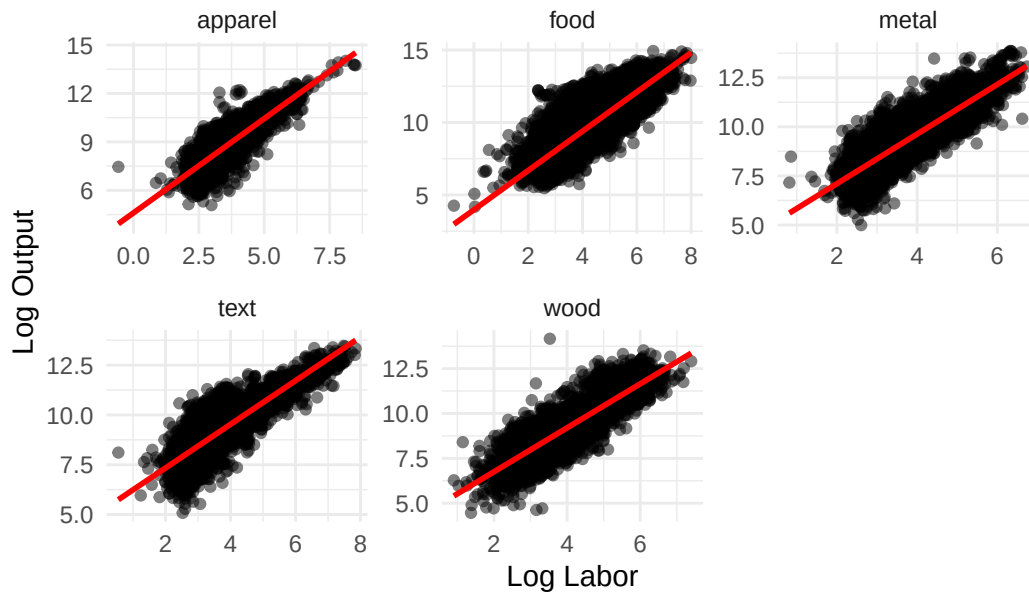
`geom\_smooth()` using formula = 'y ~ x'



```
ggplot(data_chi, aes(x = log_L, y = log_Y)) +
  geom_point(alpha = 0.5) +
  geom_smooth(method = "lm", color = "red") +
  facet_wrap(~ industry, scales = "free") +
  labs(title = "Scatter Plot of Log Output vs Log Labor (Chile)",
        x = "Log Labor",
        y = "Log Output") +
  theme_minimal()
```

`geom\_smooth()` using formula = 'y ~ x'

Scatter Plot of Log Output vs Log Labor (Chile)



Question 2

2.1

The estimating equation is as follows:

$$\log(Y_{it}) = \alpha_0 + \omega_{it} + \epsilon_{it} + \alpha_L \log(L_{it}) + \alpha_K \log(K_{it})$$

To get here, we take the log of:

$$Y_{it} = e^{\alpha_0 + \omega_{it} + \epsilon_{it}} L_{it}^{\alpha_L} K_{it}^{\alpha_K}$$

Where Y_{it} is the value added output of firm i at time t , L_{it} is labor input and K_{it} is capital input. The ω_{it} term is heterogeneous productivity shocks, varying across firms and years. As discussed in class, we can use OLS to estimate, but we will have some issues with endogeneity via ω_{it} showing up in the labor FOC.

```
# Estimate homogenous parameteres using OLS for each country
ols_col <- lm(log(Y2) ~ log_L + log_K, data = data_col)
summary(ols_col)
```

```
Call:
lm(formula = log(Y2) ~ log_L + log_K, data = data_col)

Residuals:
    Min       1Q   Median       3Q      Max
-5.9761 -0.3644 -0.0486  0.3176  6.1163

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  2.186356   0.017899  122.15  <2e-16 ***
log_L         0.830586   0.006148  135.09  <2e-16 ***
log_K         0.272688   0.003876   70.35  <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.6748 on 18080 degrees of freedom
Multiple R-squared:  0.8233,    Adjusted R-squared:  0.8233
F-statistic: 4.211e+04 on 2 and 18080 DF,  p-value: < 2.2e-16
```

```
ols_chi <- lm(log(Y2) ~ log_L + log_K, data = data_chi)
summary(ols_chi)
```

```
Call:
lm(formula = log(Y2) ~ log_L + log_K, data = data_chi)

Residuals:
    Min       1Q   Median       3Q      Max
-8.6282 -0.5256  0.0572  0.6113  3.9301

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  2.351275   0.020126  116.83  <2e-16 ***
log_L         0.887047   0.007147  124.12  <2e-16 ***
log_K         0.302505   0.003518   85.99  <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.9656 on 39662 degrees of freedom
Multiple R-squared:  0.6596,    Adjusted R-squared:  0.6596
F-statistic: 3.843e+04 on 2 and 39662 DF,  p-value: < 2.2e-16
```

```
# Estimate varying parameters across industries using OLS for each country

ols_ind_col <- lm(log(Y2) ~ industry + log_L:industry + log_K:industry,
  data = data_col)
summary(ols_ind_col)
```

Call:

```
lm(formula = log(Y2) ~ industry + log_L:industry + log_K:industry,
    data = data_col)
```

Residuals:

Min	1Q	Median	3Q	Max
-6.0224	-0.3493	-0.0412	0.3128	5.8943

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	2.510197	0.039522	63.513	< 2e-16 ***
industryfood	-0.272181	0.049287	-5.522	3.39e-08 ***
industrymetal	-0.536053	0.057780	-9.277	< 2e-16 ***
industrytext	-0.294306	0.059198	-4.972	6.70e-07 ***
industrywood	0.048779	0.088972	0.548	0.584
industryapparel:log_L	0.921453	0.013354	69.003	< 2e-16 ***
industryfood:log_L	0.754234	0.010278	73.387	< 2e-16 ***
industrymetal:log_L	0.997721	0.015540	64.205	< 2e-16 ***
industrytext:log_L	0.857699	0.015086	56.854	< 2e-16 ***
industrywood:log_L	0.974122	0.030253	32.199	< 2e-16 ***
industryapparel:log_K	0.133090	0.009339	14.251	< 2e-16 ***
industryfood:log_K	0.320732	0.006521	49.182	< 2e-16 ***
industrymetal:log_K	0.207828	0.010052	20.675	< 2e-16 ***
industrytext:log_K	0.256517	0.009788	26.206	< 2e-16 ***
industrywood:log_K	0.116640	0.017531	6.653	2.95e-11 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.6599 on 18068 degrees of freedom

Multiple R-squared: 0.8312, Adjusted R-squared: 0.831

F-statistic: 6353 on 14 and 18068 DF, p-value: < 2.2e-16

```
ols_ind_chi <- lm(log(Y2) ~ industry + log_L:industry + log_K:industry,
  data = data_chi)
summary(ols_ind_chi)
```

```
Call:
lm(formula = log(Y2) ~ industry + log_L:industry + log_K:industry,
    data = data_chi)
```

Residuals:

	Min	1Q	Median	3Q	Max
	-8.3893	-0.4762	0.0521	0.5540	5.0431

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	2.865354	0.062602	45.771	< 2e-16 ***
industryfood	-0.915420	0.068324	-13.398	< 2e-16 ***
industrymetal	0.130795	0.082198	1.591	0.1116
industrytext	0.155696	0.082572	1.886	0.0594 .
industrywood	-0.376508	0.085196	-4.419	9.93e-06 ***
industryapparel:log_L	1.054792	0.020752	50.828	< 2e-16 ***
industryfood:log_L	0.896485	0.010286	87.152	< 2e-16 ***
industrymetal:log_L	0.994447	0.018465	53.855	< 2e-16 ***
industrytext:log_L	0.874091	0.017358	50.356	< 2e-16 ***
industrywood:log_L	1.015849	0.019889	51.075	< 2e-16 ***
industryapparel:log_K	0.150469	0.012522	12.017	< 2e-16 ***
industryfood:log_K	0.333910	0.004676	71.408	< 2e-16 ***
industrymetal:log_K	0.232164	0.009310	24.938	< 2e-16 ***
industrytext:log_K	0.243121	0.009997	24.319	< 2e-16 ***
industrywood:log_K	0.174184	0.010295	16.919	< 2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.923 on 39650 degrees of freedom

Multiple R-squared: 0.689, Adjusted R-squared: 0.6889

F-statistic: 6276 on 14 and 39650 DF, p-value: < 2.2e-16

As can be seen in the regression outputs, when we add heterogeneity across industries, the estimates for labor and capital change slightly. However, we can now ascertain differences in productivity across industries via the industry coefficients.

2.2

```
form_hom <- as.formula("log(Y2) ~ log_L + log(K) + log_i")
```

```
ols_inv_col <- lm(form_hom, data = data_col)
summary(ols_inv_col)
```

Call:

```
lm(formula = form_hom, data = data_col)
```

Residuals:

Min	1Q	Median	3Q	Max
-4.8632	-0.3651	-0.0553	0.3107	6.0805

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	2.281928	0.022725	100.416	< 2e-16 ***
log_L	0.827831	0.006712	123.337	< 2e-16 ***
log(K)	0.242230	0.005785	41.871	< 2e-16 ***
log_i	0.030692	0.003856	7.959	1.87e-15 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.6355 on 14311 degrees of freedom

(3768 observations deleted due to missingness)

Multiple R-squared: 0.8361, Adjusted R-squared: 0.836

F-statistic: 2.433e+04 on 3 and 14311 DF, p-value: < 2.2e-16

```
ols_inv_chi <- lm(form_hom, data = data_chi)
summary(ols_inv_chi)
```

Call:

```
lm(formula = form_hom, data = data_chi)
```

Residuals:

Min	1Q	Median	3Q	Max
-8.6518	-0.5069	0.0514	0.5866	4.1135

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	2.520696	0.025843	97.539	<2e-16 ***
log_L	0.844182	0.008600	98.163	<2e-16 ***
log(K)	0.298895	0.004756	62.841	<2e-16 ***

```
log_i      0.016628  0.001837  9.053  <2e-16 ***
```

```
---
```

```
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
Residual standard error: 0.9253 on 25121 degrees of freedom
```

```
(14540 observations deleted due to missingness)
```

```
Multiple R-squared:  0.6895,    Adjusted R-squared:  0.6895
```

```
F-statistic: 1.859e+04 on 3 and 25121 DF,  p-value: < 2.2e-16
```

```
form_ind <- as.formula("log(Y2) ~ industry + log_L:industry + log_K:industry + log_i")
ols_inv_ind_col <- lm(form_ind, data = data_col)
summary(ols_inv_ind_col)
```

```
Call:
```

```
lm(formula = form_ind, data = data_col)
```

```
Residuals:
```

```
      Min       1Q   Median       3Q      Max
-4.7119 -0.3482 -0.0483  0.3017  5.8884
```

```
Coefficients:
```

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	2.582575	0.044736	57.729	< 2e-16 ***
industryfood	-0.240531	0.053484	-4.497	6.94e-06 ***
industrymetal	-0.509645	0.063636	-8.009	1.25e-15 ***
industrytext	-0.355354	0.065005	-5.467	4.67e-08 ***
industrywood	0.027330	0.098843	0.276	0.782
log_i	0.027448	0.003788	7.247	4.48e-13 ***
industryapparel:log_L	0.923448	0.014664	62.972	< 2e-16 ***
industryfood:log_L	0.769264	0.011107	69.261	< 2e-16 ***
industrymetal:log_L	0.983039	0.017183	57.211	< 2e-16 ***
industrytext:log_L	0.839113	0.016597	50.557	< 2e-16 ***
industrywood:log_L	0.957646	0.033943	28.214	< 2e-16 ***
industryapparel:log_K	0.106663	0.010729	9.941	< 2e-16 ***
industryfood:log_K	0.281825	0.007993	35.259	< 2e-16 ***
industrymetal:log_K	0.184796	0.011687	15.811	< 2e-16 ***
industrytext:log_K	0.246089	0.011233	21.908	< 2e-16 ***
industrywood:log_K	0.104612	0.019943	5.245	1.58e-07 ***

```
---
```

```
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Residual standard error: 0.6219 on 14299 degrees of freedom
 (3768 observations deleted due to missingness)
 Multiple R-squared: 0.8431, Adjusted R-squared: 0.843
 F-statistic: 5124 on 15 and 14299 DF, p-value: < 2.2e-16

```
ols_inv_ind_chi <- lm(form_ind, data = data_chi)
summary(ols_inv_ind_chi)
```

Call:

```
lm(formula = form_ind, data = data_chi)
```

Residuals:

Min	1Q	Median	3Q	Max
-8.4425	-0.4607	0.0403	0.5245	3.8736

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	3.122137	0.079550	39.247	< 2e-16 ***
industryfood	-1.016425	0.085203	-11.929	< 2e-16 ***
industrymetal	0.070520	0.102883	0.685	0.4931
industrytext	0.199535	0.102404	1.949	0.0514 .
industrywood	-0.504419	0.106932	-4.717	2.4e-06 ***
log_i	0.018524	0.001769	10.473	< 2e-16 ***
industryapparel:log_L	1.003792	0.026485	37.901	< 2e-16 ***
industryfood:log_L	0.842272	0.012220	68.927	< 2e-16 ***
industrymetal:log_L	0.944793	0.022022	42.903	< 2e-16 ***
industrytext:log_L	0.848320	0.020073	42.261	< 2e-16 ***
industrywood:log_L	0.977860	0.024952	39.190	< 2e-16 ***
industryapparel:log_K	0.143974	0.016572	8.688	< 2e-16 ***
industryfood:log_K	0.335670	0.006052	55.467	< 2e-16 ***
industrymetal:log_K	0.221456	0.011651	19.007	< 2e-16 ***
industrytext:log_K	0.210757	0.012327	17.097	< 2e-16 ***
industrywood:log_K	0.177565	0.013142	13.511	< 2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.8863 on 25109 degrees of freedom
 (14540 observations deleted due to missingness)
 Multiple R-squared: 0.7153, Adjusted R-squared: 0.7151
 F-statistic: 4205 on 15 and 25109 DF, p-value: < 2.2e-16

When we add investment as a regressor, we see that the coefficients on labor and capital change slightly again. The investment coefficient is positive and significant across all specifications, indicating that higher investment is associated with higher value added output. Industry level heterogeneity remains mostly significant as well.

Comparing to the results from 2.1, inclusion of investment seems to improve model fit slightly, as seen in the adjusted R-squared values, suggesting that investment is a factor worth considering when modeling firm output.

2.3

```
# First stage: estimate  $\log_y2 = \log_l + \phi(\log_k)$  s.t.  $\phi = \alpha_0 +$   
#  $\alpha_k \log_k + a(i \cdot t \cdot k)^2 + b(i \cdot t \cdot k) + c$   
  
data_col <- data_col %>%  
  arrange(id, year) %>%  
  group_by(id) %>%  
  mutate(  
    log_k_next = lead(log_K),  
    log_y2_next = lead(log_Y2)  
  ) %>%  
  ungroup()  
data_chi <- data_chi %>%  
  arrange(id, year) %>%  
  group_by(id) %>%  
  mutate(  
    log_k_next = lead(log_K),  
    log_y2_next = lead(log_Y2)  
  ) %>%  
  ungroup()  
  
# Define a function to perform the first stage regression and get phi_hat  
first_stage <- function(data) {  
  data <- data %>%  
    mutate(  
      k_sq = log_K^2,  
      it_k = as.numeric(factor(paste(id, year))) * log_K,  
      it_k_sq = it_k^2,  
      inv_sq = log_i^2,  
      k_inv = log_K * log_i,  
      k_inv_sq = k_inv^2,
```

```

      rhs = cbind(log_L, log_K, k_sq, log_i, inv_sq, k_inv, k_inv_sq)
    )
    fs_model <- lm(log_Y2 ~ rhs, data = data)
    data$phi_hat <- predict(fs_model, newdata = data)
    return(data)
  }
data_col <- first_stage(data_col)
data_chi <- first_stage(data_chi)

# Second stage: estimate  $y_{2,t+1} - \alpha_l l_{t+1} = \alpha_0 + \alpha_k k_{t+1}$ 
# +  $g(\phi_{\text{hat}_t} - \alpha_k k_t - \alpha_0) + \text{error}$ 

form_op <- as.formula("log_y2_next - log_L ~ log_k_next + I(phi_hat - log_K)")
op_col <- lm(form_op, data = data_col)
summary(op_col)

```

Call:

```
lm(formula = form_op, data = data_col)
```

Residuals:

Min	1Q	Median	3Q	Max
-5.9031	-0.3743	-0.0414	0.3349	6.6881

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	2.336745	0.030783	75.91	<2e-16 ***
log_k_next	0.162279	0.004016	40.41	<2e-16 ***
I(phi_hat - log_K)	-0.142150	0.008267	-17.19	<2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.692 on 14270 degrees of freedom

(3810 observations deleted due to missingness)

Multiple R-squared: 0.2625, Adjusted R-squared: 0.2624

F-statistic: 2540 on 2 and 14270 DF, p-value: < 2.2e-16

```

op_chi <- lm(form_op, data = data_chi)
summary(op_chi)

```

Call:

```
lm(formula = form_op, data = data_chi)
```

Residuals:

Min	1Q	Median	3Q	Max
-8.5462	-0.5352	0.0531	0.6158	4.2503

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	2.522980	0.039264	64.26	<2e-16 ***
log_k_next	0.237079	0.004571	51.87	<2e-16 ***
I(phi_hat - log_K)	-0.114318	0.009468	-12.07	<2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.9683 on 25096 degrees of freedom

(14566 observations deleted due to missingness)

Multiple R-squared: 0.2378, Adjusted R-squared: 0.2377

F-statistic: 3915 on 2 and 25096 DF, p-value: < 2.2e-16

```
# Estimate varying parameters across industries using OP for each country
# First stage already done above
data_col <- data_col %>%
  mutate(
    industry_factor = as.factor(industry)
  )
data_chi <- data_chi %>%
  mutate(
    industry_factor = as.factor(industry)
  )
form_op_ind <- as.formula("log_y2_next - log_L:industry_factor ~ industry_factor + log_k_next")
op_ind_col <- lm(form_op_ind, data = data_col)
```

Warning in log_L:industry_factor: numerical expression has 18083 elements: only the first used

Warning in log_y2_next - log_L:industry_factor: longer object length is not a multiple of shorter object length

Warning in log_K:industry_factor: numerical expression has 18083 elements: only the first used

Warning in phi_hat - log_K:industry_factor: longer object length is not a multiple of shorter object length

```
summary(op_ind_col)
```

Call:

```
lm(formula = form_op_ind, data = data_col)
```

Residuals:

Min	1Q	Median	3Q	Max
-7.9884	-1.2873	0.0136	1.2783	7.5943

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-0.722171	0.111576	-6.472	9.96e-11 ***
industry_factorfood	-1.051307	0.140270	-7.495	7.03e-14 ***
industry_factormetal	-1.100023	0.167799	-6.556	5.73e-11 ***
industry_factortext	-0.710107	0.171155	-4.149	3.36e-05 ***
industry_factorwood	-0.010793	0.253769	-0.043	0.966
I(phi_hat - log_K:industry_factor)	0.082208	0.006673	12.320	< 2e-16 ***
industry_factorapparel:log_k_next	0.545801	0.019215	28.405	< 2e-16 ***
industry_factorfood:log_k_next	0.663746	0.012672	52.381	< 2e-16 ***
industry_factormetal:log_k_next	0.676159	0.018926	35.726	< 2e-16 ***
industry_factortext:log_k_next	0.645538	0.018008	35.848	< 2e-16 ***
industry_factorwood:log_k_next	0.462320	0.036639	12.618	< 2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.677 on 14262 degrees of freedom

(3810 observations deleted due to missingness)

Multiple R-squared: 0.3865, Adjusted R-squared: 0.3861

F-statistic: 898.6 on 10 and 14262 DF, p-value: < 2.2e-16

```
op_ind_chi <- lm(form_op_ind, data = data_chi)
```

Warning in log_L:industry_factor: numerical expression has 39665 elements: only the first used

Warning in log_y2_next - log_L:industry_factor: longer object length is not a multiple of shorter object length

Warning in log_K:industry_factor: numerical expression has 39665 elements: only the first used

Warning in phi_hat - log_K:industry_factor: longer object length is not a multiple of shorter object length

```
summary(op_ind_chi)
```

Call:

```
lm(formula = form_op_ind, data = data_chi)
```

Residuals:

Min	1Q	Median	3Q	Max
-7.9528	-0.6814	0.0448	0.7592	5.7672

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)	
(Intercept)	0.590764	0.104518	5.652	1.60e-08	***
industry_factorfood	-0.565444	0.111350	-5.078	3.84e-07	***
industry_factormetal	0.430405	0.135653	3.173	0.00151	**
industry_factortext	-0.055957	0.137062	-0.408	0.68309	
industry_factorwood	-0.267367	0.140978	-1.897	0.05790	.
I(phi_hat - log_K:industry_factor)	0.071473	0.003550	20.134	< 2e-16	***
industry_factorapparel:log_k_next	0.602239	0.014838	40.588	< 2e-16	***
industry_factorfood:log_k_next	0.643781	0.005566	115.668	< 2e-16	***
industry_factormetal:log_k_next	0.581178	0.011094	52.387	< 2e-16	***
industry_factortext:log_k_next	0.587400	0.011230	52.305	< 2e-16	***
industry_factorwood:log_k_next	0.562164	0.012254	45.877	< 2e-16	***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.169 on 25088 degrees of freedom

(14566 observations deleted due to missingness)

Multiple R-squared: 0.5591, Adjusted R-squared: 0.5589

F-statistic: 3181 on 10 and 25088 DF, p-value: < 2.2e-16

2.4

```

first_stage_lp <- function(data) {
  data <- data %>%
    mutate(
      k_sq = log_K^2,
      it_k = as.numeric(factor(paste(id, year))) * log_K,
      it_k_sq = it_k^2,
      inv_sq = log_i^2,
      k_inv = log_K * log_i,
      k_inv_sq = k_inv^2,
      log_M = log(M),
      log_M_next = lead(log_M),
      m_sq = log_M^2,
      k_m = log_K * log_M,
      k_m_sq = k_m^2,
      m_inv = log_M * log_i,
      m_inv_sq = m_inv^2,
      rhs = cbind(log_L, log_K, k_sq, log_M, m_sq, log_i, inv_sq,
                  k_inv, k_inv_sq, k_m, k_m_sq, m_inv, m_inv_sq)
    )
  fs_model <- lm(log_Y2 ~ rhs, data = data)
  data$phi_hat <- predict(fs_model, newdata = data)
  return(data)
}

data_col_lp <- first_stage_lp(data_col)
data_chi_lp <- first_stage_lp(data_chi)

# Second stage for LP
form_lp <- as.formula("log_y2_next - log_L ~ log_k_next + log_M_next + I(phi_hat - log_K - 1)
lp_col <- lm(form_lp, data = data_col_lp)
summary(lp_col)

```

Call:

```
lm(formula = form_lp, data = data_col_lp)
```

Residuals:

Min	1Q	Median	3Q	Max
-4.3821	-0.3160	-0.0223	0.3076	5.0511

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	1.537257	0.023617	65.09	<2e-16 ***
log_k_next	-0.230429	0.007741	-29.77	<2e-16 ***
log_M_next	0.149754	0.005804	25.80	<2e-16 ***
I(phi_hat - log_K - log_M)	-0.306380	0.008663	-35.37	<2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.6005 on 14269 degrees of freedom

(3810 observations deleted due to missingness)

Multiple R-squared: 0.4448, Adjusted R-squared: 0.4447

F-statistic: 3811 on 3 and 14269 DF, p-value: < 2.2e-16

```
lp_chi <- lm(form_lp, data = data_chi_lp)
summary(lp_chi)
```

Call:

```
lm(formula = form_lp, data = data_chi_lp)
```

Residuals:

Min	1Q	Median	3Q	Max
-9.2076	-0.5097	0.0639	0.5920	4.4723

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	1.02117	0.03433	29.74	<2e-16 ***
log_k_next	-0.10933	0.01079	-10.13	<2e-16 ***
log_M_next	0.22070	0.00699	31.57	<2e-16 ***
I(phi_hat - log_K - log_M)	-0.25753	0.01223	-21.06	<2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.9213 on 25095 degrees of freedom

(14566 observations deleted due to missingness)

Multiple R-squared: 0.31, Adjusted R-squared: 0.3099

F-statistic: 3759 on 3 and 25095 DF, p-value: < 2.2e-16

```
# Estimate varying parameters across industries using LP for each country
data_col_lp <- data_col_lp %>%
  mutate(
    industry_factor = as.factor(industry),
```

```

    log_M_next = lead(log_M)
  )
data_chi_lp <- data_chi_lp %>%
  mutate(
    industry_factor = as.factor(industry),
    log_M_next = lead(log_M)
  )
form_lp_ind <- as.formula("log_y2_next - log_L:industry_factor ~ industry_factor + log_k_next")
lp_ind_col <- lm(form_lp_ind, data = data_col_lp)

```

Warning in log_L:industry_factor: numerical expression has 18083 elements: only the first used

Warning in log_y2_next - log_L:industry_factor: longer object length is not a multiple of shorter object length

Warning in log_K:industry_factor: numerical expression has 18083 elements: only the first used

Warning in phi_hat - log_K:industry_factor: longer object length is not a multiple of shorter object length

Warning in log_M:industry_factor: numerical expression has 18083 elements: only the first used

Warning in phi_hat - log_K:industry_factor - log_M:industry_factor: longer object length is not a multiple of shorter object length

```
summary(lp_ind_col)
```

Call:

```
lm(formula = form_lp_ind, data = data_col_lp)
```

Residuals:

Min	1Q	Median	3Q	Max
-6.7603	-1.0629	0.0005	1.0731	5.2912

Coefficients:

	Estimate	Std. Error
(Intercept)	-1.000569	0.128465
industry_factorfood	-0.772883	0.152435
industry_factormetal	-0.758896	0.175703
industry_factortext	-0.803794	0.188286
industry_factorwood	0.130881	0.292442
I(phi_hat - log_K:industry_factor - log_M:industry_factor)	0.172543	0.003406
industry_factorapparel:log_k_next	0.185429	0.022674
industry_factorfood:log_k_next	0.231630	0.016682
industry_factormetal:log_k_next	0.131470	0.027427
industry_factortext:log_k_next	0.136197	0.026698
industry_factorwood:log_k_next	0.170922	0.042629
industry_factorapparel:log_M_next	0.483433	0.024858
industry_factorfood:log_M_next	0.471494	0.017071
industry_factormetal:log_M_next	0.623566	0.029085
industry_factortext:log_M_next	0.627631	0.031215
industry_factorwood:log_M_next	0.444910	0.053687

	t value	Pr(> t)
(Intercept)	-7.789	7.24e-15 ***
industry_factorfood	-5.070	4.02e-07 ***
industry_factormetal	-4.319	1.58e-05 ***
industry_factortext	-4.269	1.98e-05 ***
industry_factorwood	0.448	0.654
I(phi_hat - log_K:industry_factor - log_M:industry_factor)	50.665	< 2e-16 ***
industry_factorapparel:log_k_next	8.178	3.12e-16 ***
industry_factorfood:log_k_next	13.885	< 2e-16 ***
industry_factormetal:log_k_next	4.794	1.66e-06 ***
industry_factortext:log_k_next	5.101	3.42e-07 ***
industry_factorwood:log_k_next	4.010	6.11e-05 ***
industry_factorapparel:log_M_next	19.448	< 2e-16 ***
industry_factorfood:log_M_next	27.619	< 2e-16 ***
industry_factormetal:log_M_next	21.439	< 2e-16 ***
industry_factortext:log_M_next	20.107	< 2e-16 ***
industry_factorwood:log_M_next	8.287	< 2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.435 on 14257 degrees of freedom
(3810 observations deleted due to missingness)

Multiple R-squared: 0.5509, Adjusted R-squared: 0.5504

F-statistic: 1166 on 15 and 14257 DF, p-value: < 2.2e-16

```
lp_ind_chi <- lm(form_lp_ind, data = data_chi_lp)
```

Warning in log_L:industry_factor: numerical expression has 39665 elements: only the first used

Warning in log_y2_next - log_L:industry_factor: longer object length is not a multiple of shorter object length

Warning in log_K:industry_factor: numerical expression has 39665 elements: only the first used

Warning in phi_hat - log_K:industry_factor: longer object length is not a multiple of shorter object length

Warning in log_M:industry_factor: numerical expression has 39665 elements: only the first used

Warning in phi_hat - log_K:industry_factor - log_M:industry_factor: longer object length is not a multiple of shorter object length

```
summary(lp_ind_chi)
```

Call:

```
lm(formula = form_lp_ind, data = data_chi_lp)
```

Residuals:

Min	1Q	Median	3Q	Max
-8.7065	-0.5619	0.0513	0.6629	4.3827

Coefficients:

	Estimate	Std. Error
(Intercept)	-1.698662	0.122759
industry_factorfood	-0.732189	0.131039
industry_factormetal	0.438701	0.156775
industry_factortext	-0.048067	0.160793
industry_factorwood	-0.538117	0.164958
I(phi_hat - log_K:industry_factor - log_M:industry_factor)	0.048840	0.002343
industry_factorapparel:log_k_next	0.243267	0.017603

industry_factorfood:log_k_next	0.221859	0.007706
industry_factormetal:log_k_next	0.213917	0.013906
industry_factortext:log_k_next	0.258298	0.014067
industry_factorwood:log_k_next	0.182565	0.015184
industry_factorapparel:log_M_next	0.603509	0.020068
industry_factorfood:log_M_next	0.662323	0.009760
industry_factormetal:log_M_next	0.624276	0.017088
industry_factortext:log_M_next	0.590802	0.018072
industry_factorwood:log_M_next	0.673808	0.019216
	t value	Pr(> t)
(Intercept)	-13.837	< 2e-16 ***
industry_factorfood	-5.588	2.33e-08 ***
industry_factormetal	2.798	0.00514 **
industry_factortext	-0.299	0.76499
industry_factorwood	-3.262	0.00111 **
I(phi_hat - log_K:industry_factor - log_M:industry_factor)	20.841	< 2e-16 ***
industry_factorapparel:log_k_next	13.820	< 2e-16 ***
industry_factorfood:log_k_next	28.790	< 2e-16 ***
industry_factormetal:log_k_next	15.383	< 2e-16 ***
industry_factortext:log_k_next	18.362	< 2e-16 ***
industry_factorwood:log_k_next	12.023	< 2e-16 ***
industry_factorapparel:log_M_next	30.073	< 2e-16 ***
industry_factorfood:log_M_next	67.860	< 2e-16 ***
industry_factormetal:log_M_next	36.532	< 2e-16 ***
industry_factortext:log_M_next	32.692	< 2e-16 ***
industry_factorwood:log_M_next	35.065	< 2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.9898 on 25083 degrees of freedom

(14566 observations deleted due to missingness)

Multiple R-squared: 0.6843, Adjusted R-squared: 0.6841

F-statistic: 3624 on 15 and 25083 DF, p-value: < 2.2e-16

2.5

```
# ACF Estimation for each country
acf_fs <- function(data) {
  data <- data %>%
    mutate(
      lag_log_L = lag(log_L),
```

```

    lag_log_K = lag(log_K),
    t2_lag_log_L = lag(lag_log_L),
    t2_lag_log_K = lag(lag_log_K),
    k_sq = log_K^2,
    it_k = as.numeric(factor(paste(id, year))) * log_K,
    it_k_sq = it_k^2,
    l_sq = log_L^2,
    it_l = as.numeric(factor(paste(id, year))) * log_L,
    it_l_sq = it_l^2,
    l_k = log_L * log_K,
    l_k_sq = l_k^2,
    rhs = cbind(lag_log_L, lag_log_K, t2_lag_log_L, t2_lag_log_K,
                k_sq, l_sq, it_k, it_k_sq, it_l, it_l_sq, l_k, l_k_sq)
  )
  fs_model <- lm(log_Y2 ~ rhs, data = data)
  data$phi_hat <- predict(fs_model, newdata = data)
  return(data)
}

data_col_acf <- acf_fs(data_col)
data_chi_acf <- acf_fs(data_chi)
# Second stage for ACF
form_acf <- as.formula("log_Y2 ~ log_L + log_K + I(phi_hat)")
acf_col <- lm(form_acf, data = data_col_acf)
summary(acf_col)

```

Call:

```
lm(formula = form_acf, data = data_col_acf)
```

Residuals:

Min	1Q	Median	3Q	Max
-6.0051	-0.3314	-0.0358	0.2999	5.9090

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	0.283969	0.058938	4.818	1.46e-06 ***
log_L	0.116297	0.021973	5.293	1.22e-07 ***
log_K	0.033723	0.008012	4.209	2.58e-05 ***
I(phi_hat)	0.867214	0.025675	33.776	< 2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.6546 on 18077 degrees of freedom
 (2 observations deleted due to missingness)
 Multiple R-squared: 0.8337, Adjusted R-squared: 0.8337
 F-statistic: 3.021e+04 on 3 and 18077 DF, p-value: < 2.2e-16

```
acf_chi <- lm(form_acf, data = data_chi_acf)
summary(acf_chi)
```

Call:

```
lm(formula = form_acf, data = data_chi_acf)
```

Residuals:

Min	1Q	Median	3Q	Max
-8.6796	-0.5132	0.0640	0.6027	3.9449

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	0.014412	0.082378	0.175	0.861
log_L	0.018050	0.030552	0.591	0.555
log_K	-0.004735	0.011072	-0.428	0.669
I(phi_hat)	0.993972	0.033999	29.235	<2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.9553 on 39659 degrees of freedom
 (2 observations deleted due to missingness)
 Multiple R-squared: 0.6668, Adjusted R-squared: 0.6668
 F-statistic: 2.645e+04 on 3 and 39659 DF, p-value: < 2.2e-16

```
# Estimate varying parameters across industries using ACF for each country
data_col_acf <- data_col_acf %>%
  mutate(
    industry_factor = as.factor(industry)
  )
data_chi_acf <- data_chi_acf %>%
  mutate(
    industry_factor = as.factor(industry)
  )
form_acf_ind <- as.formula("log_Y2 ~ industry_factor + log_L:industry_factor + log_K:industry_factor")
```

```
acf_ind_col <- lm(form_acf_ind, data = data_col_acf)
summary(acf_ind_col)
```

Call:

```
lm(formula = form_acf_ind, data = data_col_acf)
```

Residuals:

Min	1Q	Median	3Q	Max
-6.0574	-0.3271	-0.0363	0.2960	5.8703

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	1.48274	0.15427	9.611	< 2e-16 ***
industry_factorfood	-1.17760	0.18041	-6.527	6.88e-11 ***
industry_factormetal	-1.27686	0.21761	-5.868	4.49e-09 ***
industry_factortext	-1.00531	0.20920	-4.805	1.56e-06 ***
industry_factorwood	0.47531	0.30113	1.578	0.114483
industry_factorapparel:log_L	0.54222	0.05665	9.571	< 2e-16 ***
industry_factorfood:log_L	0.03206	0.03474	0.923	0.356099
industry_factormetal:log_L	0.30837	0.05960	5.174	2.32e-07 ***
industry_factortext:log_L	0.20877	0.05235	3.988	6.69e-05 ***
industry_factorwood:log_L	0.75844	0.09332	8.127	4.67e-16 ***
industry_factorapparel:log_K	0.04991	0.01515	3.294	0.000991 ***
industry_factorfood:log_K	0.05851	0.01365	4.285	1.84e-05 ***
industry_factormetal:log_K	-0.01099	0.02077	-0.529	0.596702
industry_factortext:log_K	0.02316	0.02044	1.133	0.257281
industry_factorwood:log_K	0.05325	0.03115	1.709	0.087439 .
industry_factorapparel:I(phi_hat)	0.42894	0.06235	6.879	6.21e-12 ***
industry_factorfood:I(phi_hat)	0.89309	0.04112	21.721	< 2e-16 ***
industry_factormetal:I(phi_hat)	0.82053	0.06860	11.962	< 2e-16 ***
industry_factortext:I(phi_hat)	0.80251	0.06212	12.919	< 2e-16 ***
industry_factorwood:I(phi_hat)	0.25904	0.10630	2.437	0.014819 *

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.6452 on 18061 degrees of freedom

(2 observations deleted due to missingness)

Multiple R-squared: 0.8386, Adjusted R-squared: 0.8384

F-statistic: 4938 on 19 and 18061 DF, p-value: < 2.2e-16

```
acf_ind_chi <- lm(form_acf_ind, data = data_chi_acf)
summary(acf_ind_chi)
```

Call:

```
lm(formula = form_acf_ind, data = data_chi_acf)
```

Residuals:

Min	1Q	Median	3Q	Max
-8.5934	-0.4639	0.0593	0.5532	4.0975

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)	
(Intercept)	1.575895	0.300133	5.251	1.52e-07	***
industry_factorfood	-2.431524	0.319932	-7.600	3.02e-14	***
industry_factormetal	0.785799	0.362118	2.170	0.03001	*
industry_factortext	0.539087	0.362589	1.487	0.13708	
industry_factorwood	-1.612979	0.398402	-4.049	5.16e-05	***
industry_factorapparel:log_L	0.564594	0.113514	4.974	6.59e-07	***
industry_factorfood:log_L	-0.129194	0.040569	-3.185	0.00145	**
industry_factormetal:log_L	0.741621	0.080070	9.262	< 2e-16	***
industry_factortext:log_L	0.538974	0.074624	7.223	5.20e-13	***
industry_factorwood:log_L	0.079225	0.096830	0.818	0.41325	
industry_factorapparel:log_K	-0.007223	0.037992	-0.190	0.84921	
industry_factorfood:log_K	-0.020055	0.014325	-1.400	0.16153	
industry_factormetal:log_K	0.137531	0.030599	4.495	6.99e-06	***
industry_factortext:log_K	0.116859	0.029095	4.017	5.92e-05	***
industry_factorwood:log_K	-0.159946	0.035323	-4.528	5.97e-06	***
industry_factorapparel:I(phi_hat)	0.544098	0.123915	4.391	1.13e-05	***
industry_factorfood:I(phi_hat)	1.171934	0.044877	26.115	< 2e-16	***
industry_factormetal:I(phi_hat)	0.287279	0.088580	3.243	0.00118	**
industry_factortext:I(phi_hat)	0.390735	0.084672	4.615	3.95e-06	***
industry_factorwood:I(phi_hat)	1.078384	0.109157	9.879	< 2e-16	***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.9135 on 39643 degrees of freedom

(2 observations deleted due to missingness)

Multiple R-squared: 0.6954, Adjusted R-squared: 0.6953

F-statistic: 4764 on 19 and 39643 DF, p-value: < 2.2e-16

Question 3

3.1